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Roll No

IT-227-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Communication System

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions
ii) All questions carry equal marks.

1. a) Comment on frequency deviation, deviation ratio and Carson's rule.
b) With the help of suitable diagrams explain direct and indirect methods of FM generation.
2. Draw a block diagram of communication system and explain it briefly.
3. a) What is Double Side Band Suppressed Carrier (DSB-SC) modulation? Explain the basic principle of DSB-SC modulation with suitable sketch.
b) When the modulation percentage is 75%, an AM transmitter radiates 10kW power? How much of this is carrier Power?
4. Give the difference between Frequency Modulation(FM) and Phase Modulation (PM).

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5. a) Write and explain time and frequency shifting properties of Fourier transform.
b) Write and explain the shifting property of Delta function. Evaluate the following integrals.

i) $\int_3^5 [t^3 + 4t + 2] \delta(t-1) dt$

ii) $\int_{-\infty}^{\infty} [t^4 - 2] \delta(1-t) dt$

6. a) Which one of the following system is a causal system?

i) $y(t) = \sin[u(t+3)]$

ii) $y(t) = 5u(t) + 3u(t-1)$

iii) $y(t) = 5u(t) + 3u(t+1)$

- b) What is a Coaxial cable? Which is the most common cable used for video signals?

7. a) With reference to radio receivers. Define the following terms

i) Sensitivity

ii) Selectivity

iii) Fidelity

- b) With a block diagram, explain the working of high power AM transmitter.

8. Draw and explain any two of the following signals.

i) Sinusoidal

ii) Unit step

iii) Ramp

iv) Signum

v) Rectangular pulse

vi) Impulse (delta)

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